

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 9: Matrices

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Multiplication of matrices

1. If $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, find A^2 . (a) $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ **ICSE 2025, 1 mark**

SOLUTION

$$A^2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0+1 & 0+0 \\ 0+0 & 1+0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Answer: (c)

2. Given $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$ and $C = \begin{bmatrix} -5 & 1 \\ 7 & -4 \end{bmatrix}$, find (a) $A + C$ (b) $B(A + C)$ (c) $5B$ (d) $B(A + C) - 5B$. **ICSE 2025, 4 marks**

SOLUTION

$$(a) A + C = \begin{bmatrix} 1-5 & 2+1 \\ 3+7 & 4-4 \end{bmatrix} = \begin{bmatrix} -4 & 3 \\ 10 & 0 \end{bmatrix}$$

$$(b) B(A + C) = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} -4 & 3 \\ 10 & 0 \end{bmatrix} = \begin{bmatrix} -8+10 & 6+0 \\ -16+20 & 12+0 \end{bmatrix} = \begin{bmatrix} 2 & 6 \\ 4 & 12 \end{bmatrix}$$

$$(c) 5B = \begin{bmatrix} 10 & 5 \\ 20 & 10 \end{bmatrix}$$

$$(d) B(A + C) - 5B = \begin{bmatrix} 2-10 & 6-5 \\ 4-20 & 12-10 \end{bmatrix}$$

$$\text{Answer: (d)} \begin{bmatrix} -8 & 1 \\ -16 & 2 \end{bmatrix}$$

3. If $A = \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix}$ and $A^2 = \begin{bmatrix} 4 & x \\ 0 & 4 \end{bmatrix}$, find x . (a) 2 (b) 4 (c) 8 (d) 10 **ICSE 2024, 1 mark**

SOLUTION

$$A^2 = \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 4+0 & 4+4 \\ 0+0 & 0+4 \end{bmatrix} = \begin{bmatrix} 4 & 8 \\ 0 & 4 \end{bmatrix}$$

Answer: (c)

4. Given $A = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$, $C = \begin{bmatrix} 4 & 1 \\ 1 & 5 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, find $A(B + C) - 14I$. **ICSE 2023, 3 marks**

SOLUTION

$$B + C = \begin{bmatrix} 5 & 3 \\ 3 & 9 \end{bmatrix}$$

$$A(B + C) = \begin{bmatrix} 5 + 9 & 3 + 27 \\ 10 + 12 & 6 + 36 \end{bmatrix} = \begin{bmatrix} 14 & 30 \\ 22 & 42 \end{bmatrix}$$

$$14I = \begin{bmatrix} 14 & 0 \\ 0 & 14 \end{bmatrix}$$

$$\text{Answer: } \begin{bmatrix} 0 & 30 \\ 22 & 28 \end{bmatrix}$$

5. If $A = \begin{bmatrix} 3 & 0 \\ 5 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -4 & 2 \\ 1 & 0 \end{bmatrix}$, find $A^2 - 2AB + B^2$. **ICSE 2020, 3 marks**

SOLUTION

$$A^2 = \begin{bmatrix} 9 + 0 & 0 + 0 \\ 15 + 5 & 0 + 1 \end{bmatrix} = \begin{bmatrix} 9 & 0 \\ 20 & 1 \end{bmatrix}$$

$$AB = \begin{bmatrix} -12 + 0 & 6 + 0 \\ -20 + 1 & 10 + 0 \end{bmatrix} = \begin{bmatrix} -12 & 6 \\ -19 & 10 \end{bmatrix}, \text{ so } 2AB = \begin{bmatrix} -24 & 12 \\ -38 & 20 \end{bmatrix}$$

$$B^2 = \begin{bmatrix} 16 + 2 & -8 + 0 \\ -4 + 0 & 2 + 0 \end{bmatrix} = \begin{bmatrix} 18 & -8 \\ -4 & 2 \end{bmatrix}$$

$$A^2 - 2AB + B^2 = \begin{bmatrix} 9 + 24 + 18 & 0 - 12 - 8 \\ 20 + 38 - 4 & 1 - 20 + 2 \end{bmatrix}$$

$$\text{Answer: } \begin{bmatrix} 51 & -20 \\ 54 & -17 \end{bmatrix}$$

6. If $A = \begin{bmatrix} 2 & 3 \\ 5 & 7 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 4 \\ -1 & 7 \end{bmatrix}$ and $C = \begin{bmatrix} 1 & 0 \\ -1 & 4 \end{bmatrix}$, find $AC + B^2 - 10C$. **ICSE 2018, 3 marks**

SOLUTION

$$AC = \begin{bmatrix} 2 - 3 & 0 + 12 \\ 5 - 7 & 0 + 28 \end{bmatrix} = \begin{bmatrix} -1 & 12 \\ -2 & 28 \end{bmatrix}$$

$$B^2 = \begin{bmatrix} 0 - 4 & 0 + 28 \\ 0 - 7 & -4 + 49 \end{bmatrix} = \begin{bmatrix} -4 & 28 \\ -7 & 45 \end{bmatrix} \text{ and } 10C = \begin{bmatrix} 10 & 0 \\ -10 & 40 \end{bmatrix}$$

$$AC + B^2 - 10C = \begin{bmatrix} -1 - 4 - 10 & 12 + 28 - 0 \\ -2 - 7 + 10 & 28 + 45 - 40 \end{bmatrix}$$

$$\text{Answer: } \begin{bmatrix} -15 & 40 \\ 1 & 33 \end{bmatrix}$$

7. $A = \begin{bmatrix} 1 & 3 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} -2 & 1 \\ -3 & 2 \end{bmatrix}$ and $A^2 - 5B^2 = 5C$. Find the matrix C . **ICSE 2017, 4 marks**

SOLUTION

$$A^2 = \begin{bmatrix} 1+9 & 3+12 \\ 3+12 & 9+16 \end{bmatrix} = \begin{bmatrix} 10 & 15 \\ 15 & 25 \end{bmatrix}$$

$$B^2 = \begin{bmatrix} 4-3 & -2+2 \\ 6-6 & -3+4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$5C = A^2 - 5B^2 = \begin{bmatrix} 5 & 15 \\ 15 & 20 \end{bmatrix}$$

Answer: $C = \begin{bmatrix} 1 & 3 \\ 3 & 4 \end{bmatrix}$

2 Solving matrix equations

8. $A = \begin{bmatrix} x & 0 \\ 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 4 & 0 \\ y & 1 \end{bmatrix}$ and $C = \begin{bmatrix} 4 & 0 \\ x & 1 \end{bmatrix}$. Find x and y if $AB = C$. **ICSE 2024, 4 marks**

SOLUTION

$$AB = \begin{bmatrix} 4x+0 & 0+0 \\ 4+y & 0+1 \end{bmatrix} = \begin{bmatrix} 4x & 0 \\ 4+y & 1 \end{bmatrix}$$

Compare with C : $4x = 4$, so $x = 1$. Then $4 + y = x = 1$, so $y = -3$.

Answer: $x = 1, y = -3$

9. If $\begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ -8 \end{bmatrix}$, find x and y . (a) 1, -2 (b) -2, 1 (c) 1, 2 (d) -2, -1 **ICSE 2023, 1 mark**

SOLUTION

Multiply out: $\begin{bmatrix} 2x \\ 4y \end{bmatrix} = \begin{bmatrix} 2 \\ -8 \end{bmatrix}$, so $x = 1$ and $y = -2$.

Answer: (a)

10. $A = \begin{bmatrix} x & 3 \\ y & 3 \end{bmatrix}$ and $A^2 = 3I$, where I is the identity matrix of order 2. Find x and y . **ICSE 2020, 3 marks**

SOLUTION

$$A^2 = \begin{bmatrix} x^2+3y & 3x+9 \\ xy+3y & 3y+9 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$$

$3x + 9 = 0$ gives $x = -3$, and $3y + 9 = 3$ gives $y = -2$.

Check: $x^2 + 3y = 9 - 6 = 3$ and $xy + 3y = 6 - 6 = 0$.

Answer: $x = -3, y = -2$

- 11.** Given $\begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix} M = 6I$, where I is the unit matrix of order 2×2 , (a) state the order of M (b) find M . **ICSE 2019, 4 marks**

SOLUTION

(a) The product is a 2×2 matrix and the first matrix is 2×2 , so M is 2×2 .

(b) Let $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Then $\begin{bmatrix} 4a + 2c & 4b + 2d \\ -a + c & -b + d \end{bmatrix} = \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix}$.

$-a + c = 0$ gives $c = a$, and then $6a = 6$, so $a = 1$ and $c = 1$.

$-b + d = 6$ gives $d = b + 6$, and then $4b + 2b + 12 = 0$, so $b = -2$ and $d = 4$.

Answer: $M = \begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix}$

- 12.** Given $B = \begin{bmatrix} 1 & 1 \\ 8 & 3 \end{bmatrix}$, find $X = B^2 - 4B$. Then solve $X \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 5 \\ 50 \end{bmatrix}$ for a and b . **ICSE 2017, 4 marks**

SOLUTION

$$B^2 = \begin{bmatrix} 1+8 & 1+3 \\ 8+24 & 8+9 \end{bmatrix} = \begin{bmatrix} 9 & 4 \\ 32 & 17 \end{bmatrix} \text{ and } 4B = \begin{bmatrix} 4 & 4 \\ 32 & 12 \end{bmatrix}$$

$$X = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$$

$$X \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 5a \\ 5b \end{bmatrix} = \begin{bmatrix} 5 \\ 50 \end{bmatrix}$$

Answer: $X = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}; a = 1, b = 10$

3 Addition, subtraction and multiplication by a number

- 13.** Simplify $\sin A \begin{bmatrix} \sin A & -\cos A \\ \cos A & \sin A \end{bmatrix} + \cos A \begin{bmatrix} \cos A & \sin A \\ -\sin A & \cos A \end{bmatrix}$. **ICSE 2019, 3 marks**

SOLUTION

Multiply each matrix by its number, then add the matching entries.

$$= \begin{bmatrix} \sin^2 A + \cos^2 A & -\sin A \cos A + \sin A \cos A \\ \sin A \cos A - \sin A \cos A & \sin^2 A + \cos^2 A \end{bmatrix}$$

Answer: $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

- 14.** Find x and y if $2 \begin{bmatrix} x & 7 \\ 9 & y-5 \end{bmatrix} + \begin{bmatrix} 6 & -7 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} 10 & 7 \\ 22 & 15 \end{bmatrix}$. **ICSE 2018, 3 marks**

SOLUTION

$$\text{Left side} = \begin{bmatrix} 2x+6 & 14-7 \\ 18+4 & 2y-10+5 \end{bmatrix} = \begin{bmatrix} 2x+6 & 7 \\ 22 & 2y-5 \end{bmatrix}$$

$2x + 6 = 10$ gives $x = 2$, and $2y - 5 = 15$ gives $y = 10$.

Answer: $x = 2, y = 10$

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